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DUBLIN
TECHNOLOGICAL UNIVERSITY DUBLIN

Secure Documents Vault Mobile Application

Danyil Tymchyk, Illia Stefanovskyi, Artem Surszenko

14/05/2026

GitHub:

- Danyil Tymchuk
- Illia Stefanovskyi
- Artem Surzhenko

DanyilT/
SecDocVault

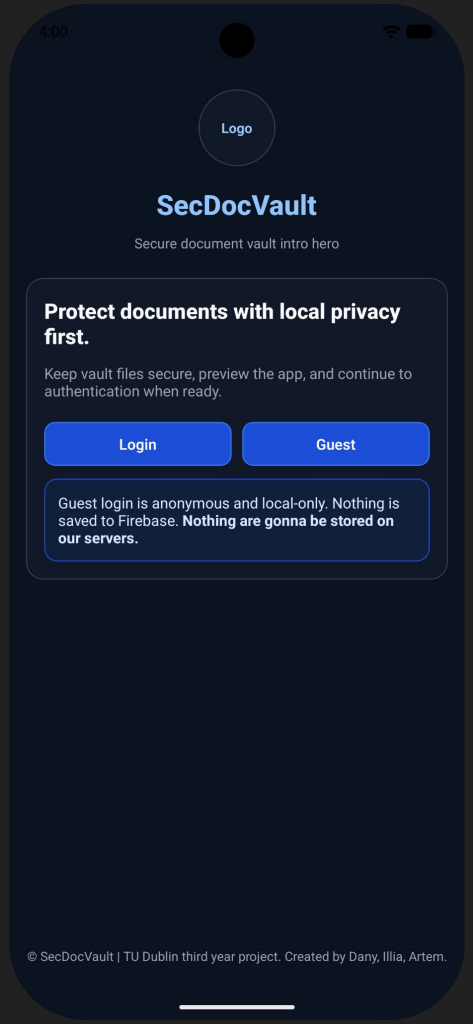


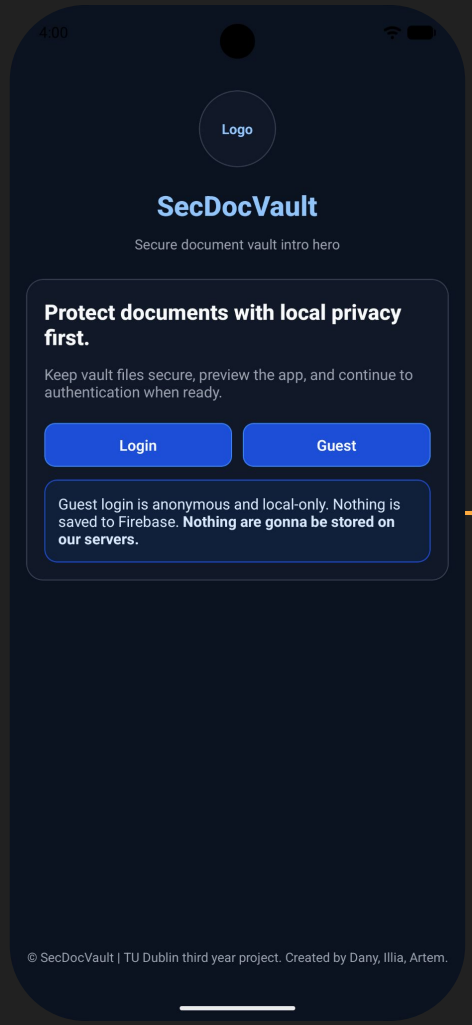
Secure Documents Vault Mobile Application (third year team project)

3 Contributors | 0 Issues | 1 Star | 0 Forks

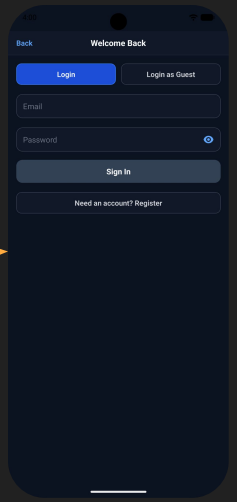
https://github.com/DanyilT/SecDocVault/releases/tag/v0.1.1

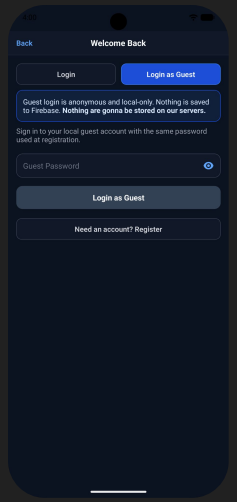
App Walkthrough


4:00
Logo
SecDocVault
Secure document vault intro hero
Protect documents with local privacy first.
Keep vault files secure, preview the app, and continue to authentication when ready.
Login
Guest
Guest login is anonymous and local-only. Nothing is saved to Firebase. Nothing are gonna be stored on our servers.
© SecDocVault | TU Dublin third year project. Created by Dany, Illia, Artem.

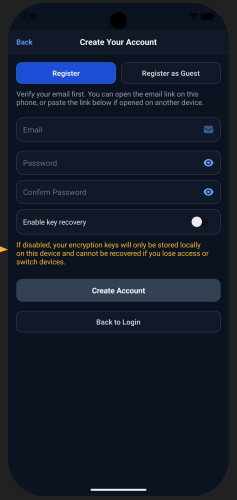


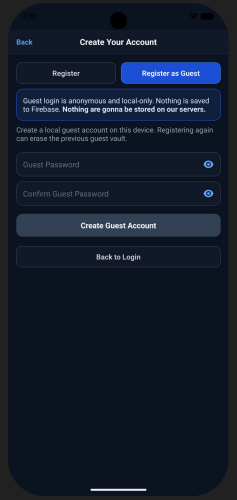
login



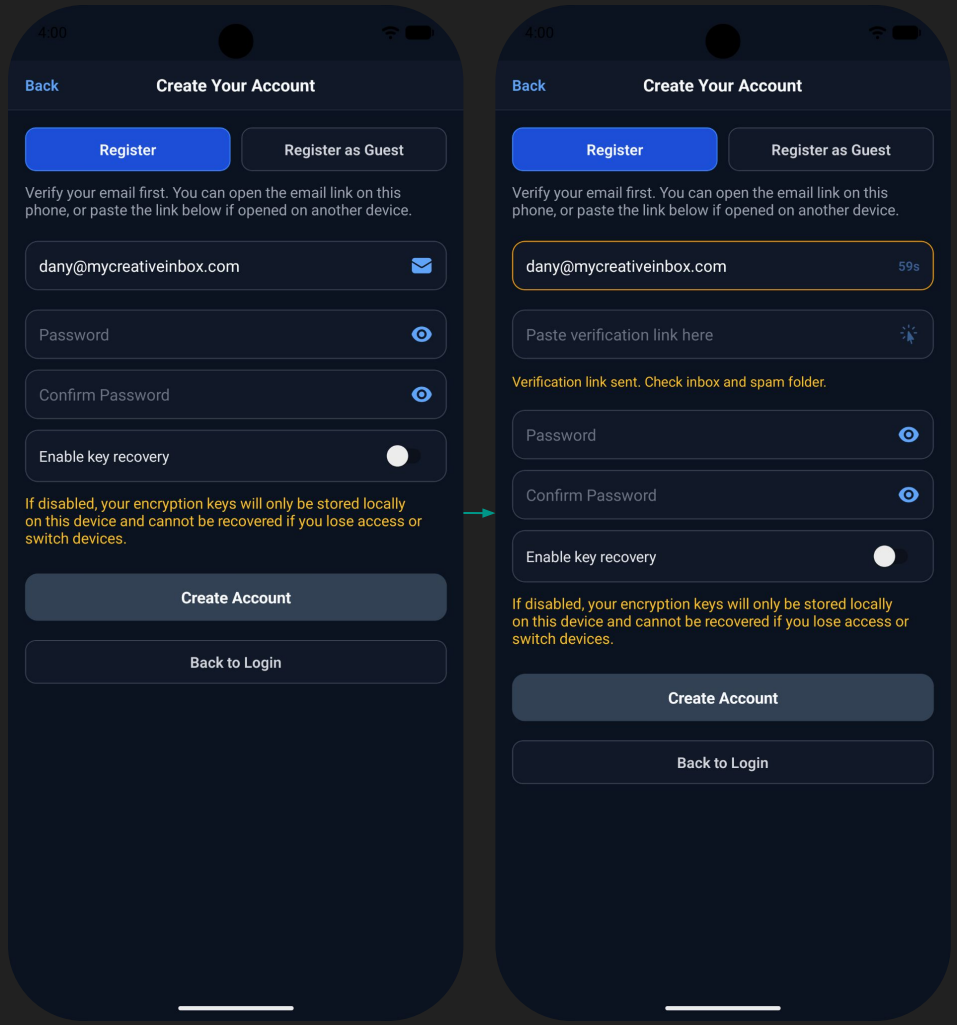


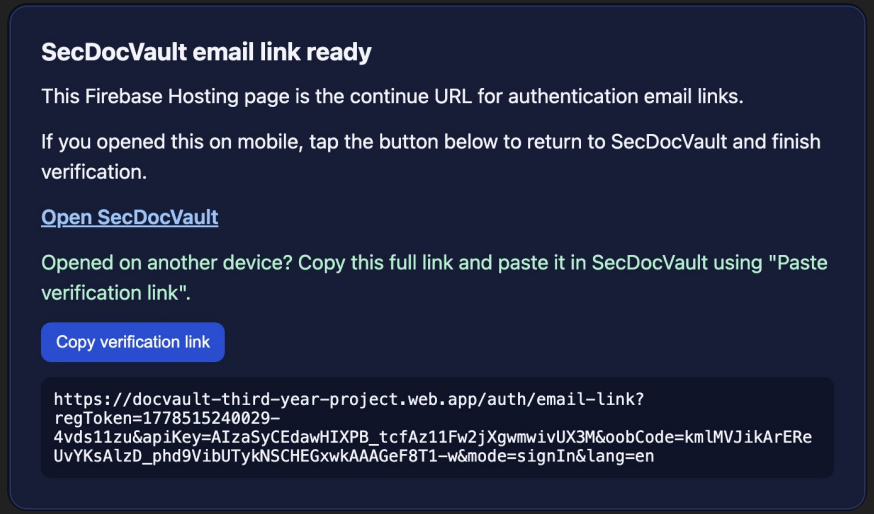
register





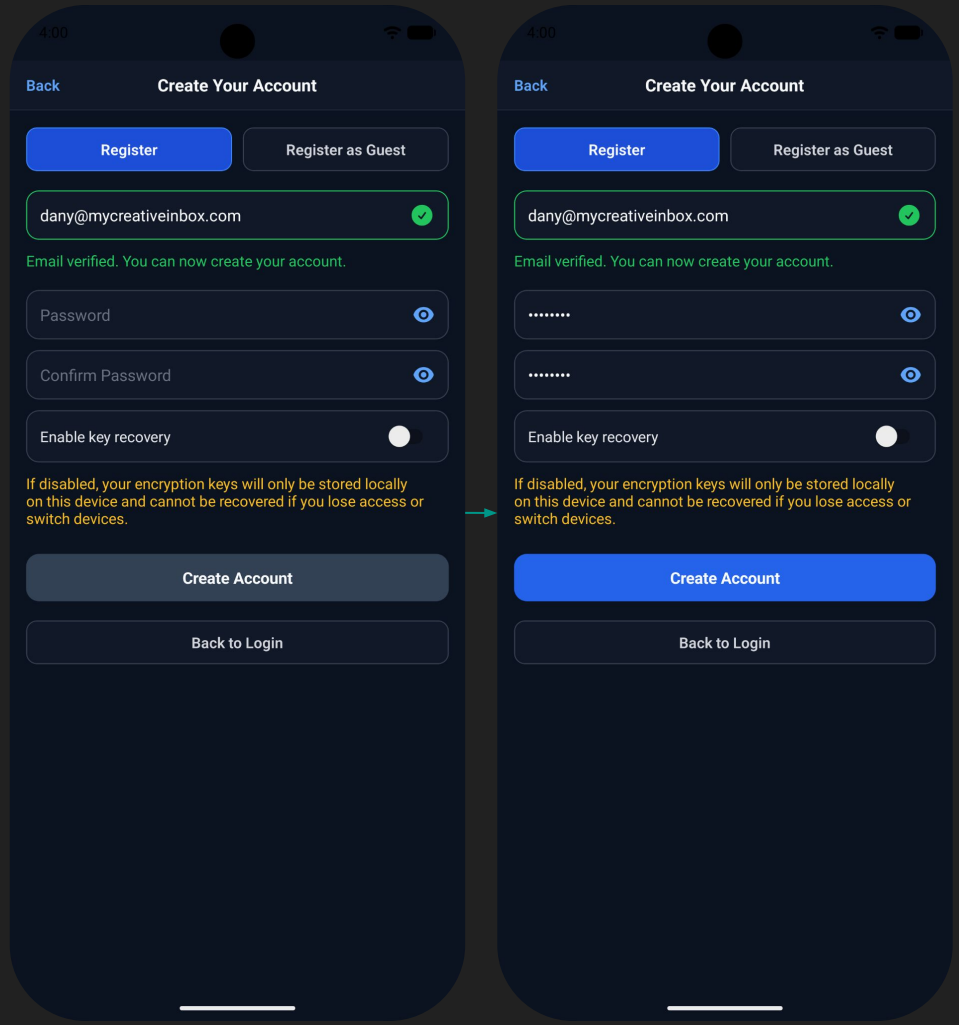
Lets dive into Registration

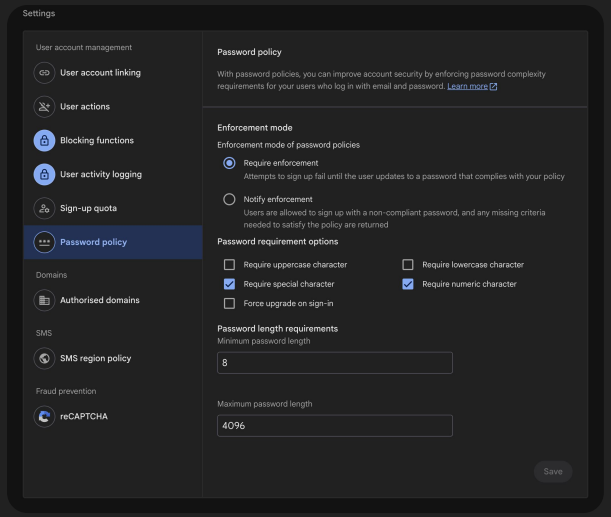




Verify user's email at the start

to ensure user used own email and not someone else (spoofing)

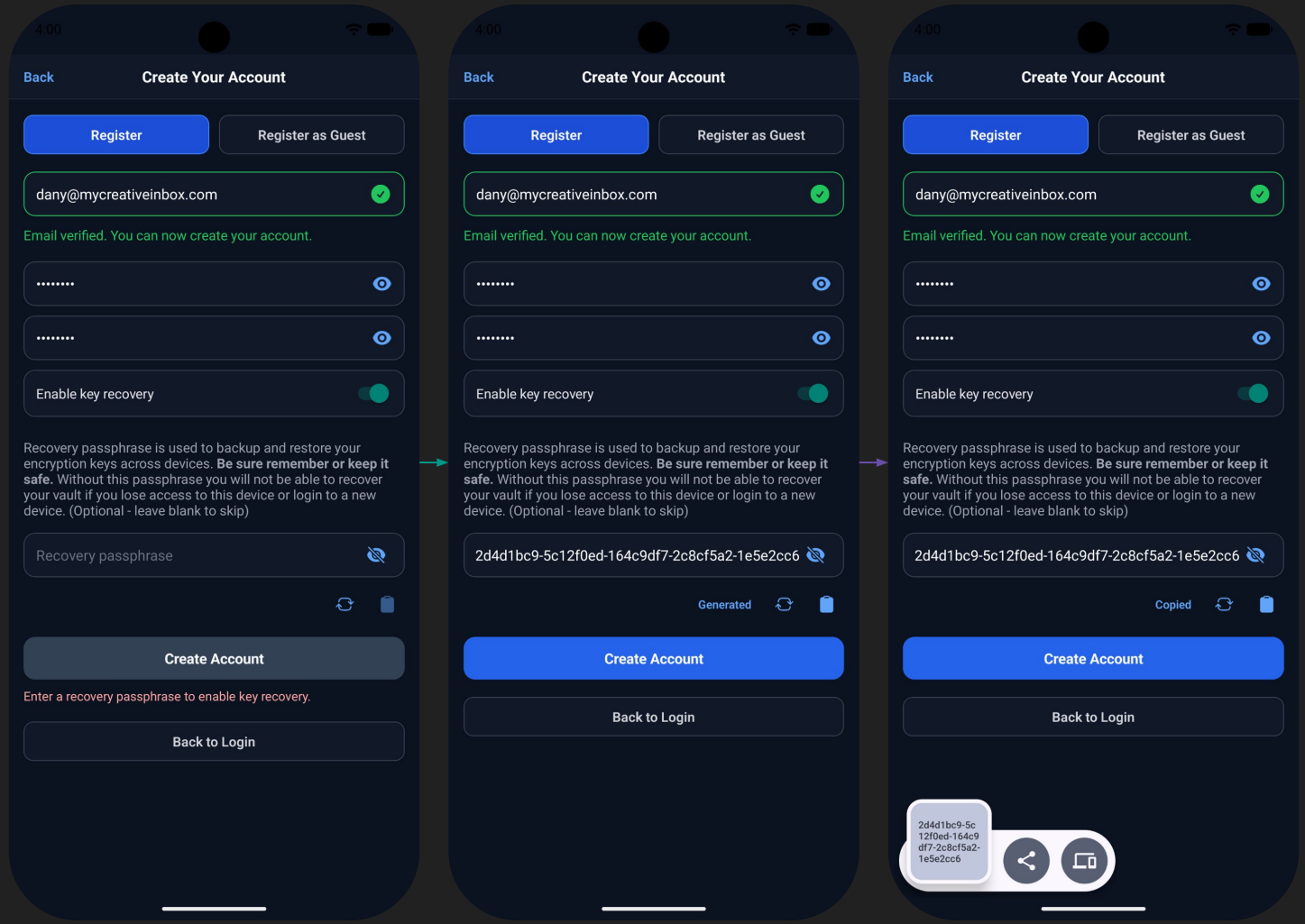




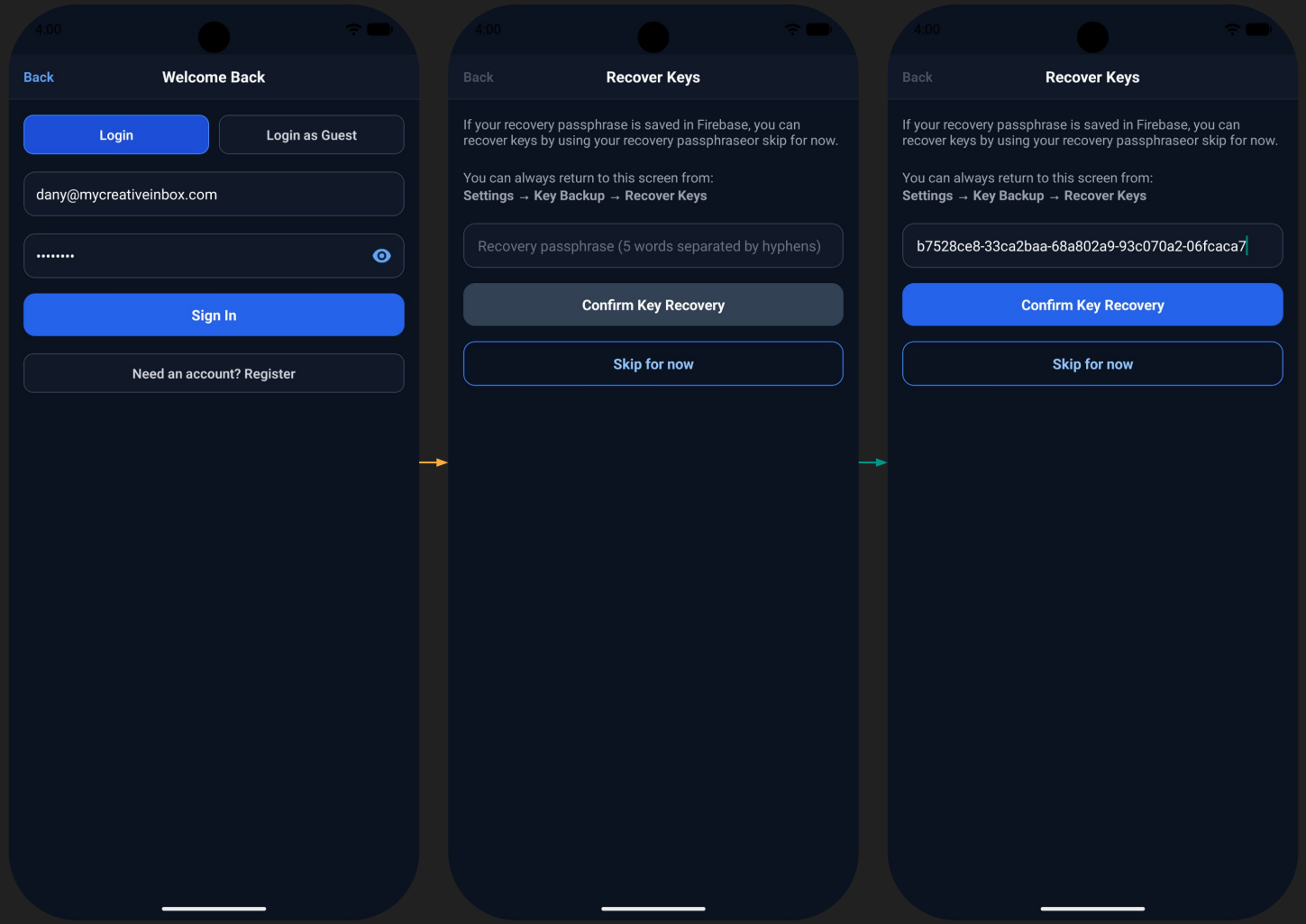
* password policy

Create a strong password

Enable Keys Recovery



Lets dive into Login

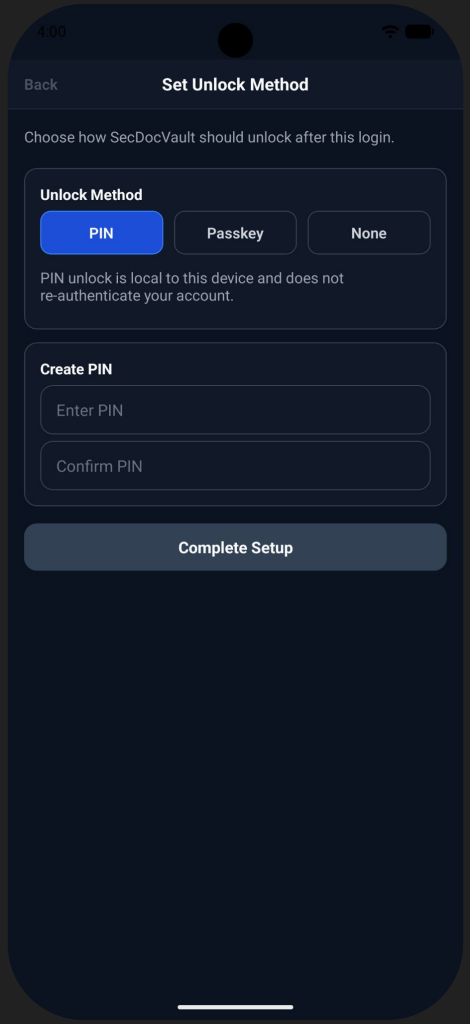


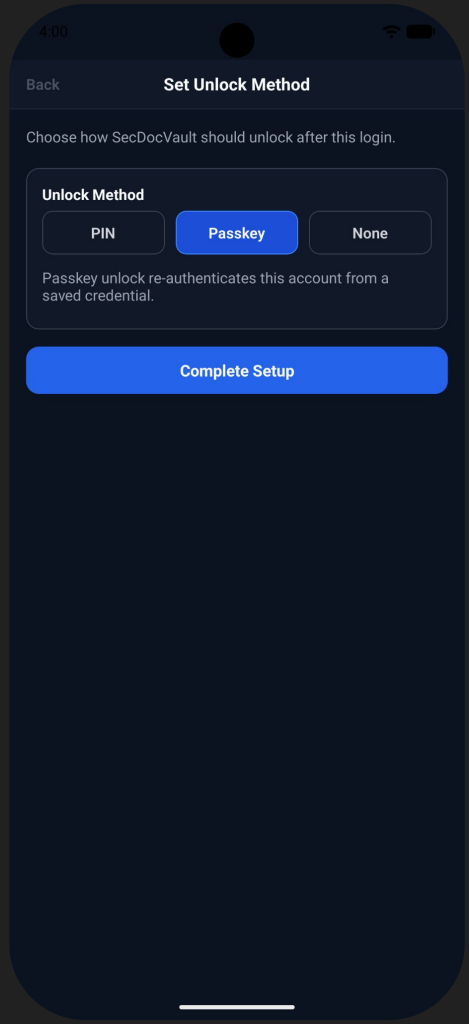
If a user had setup a keys recovery, this user can restore all saved keys now (or later in settings)

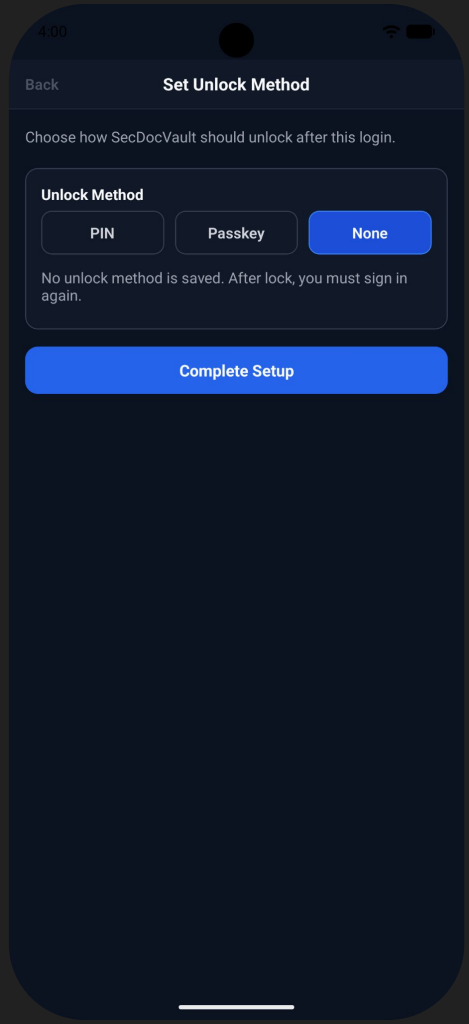
We will talk more about Keys Recovery in a couple of slides

App Lock Screen — Setup Unlock Method

can be changed in settings



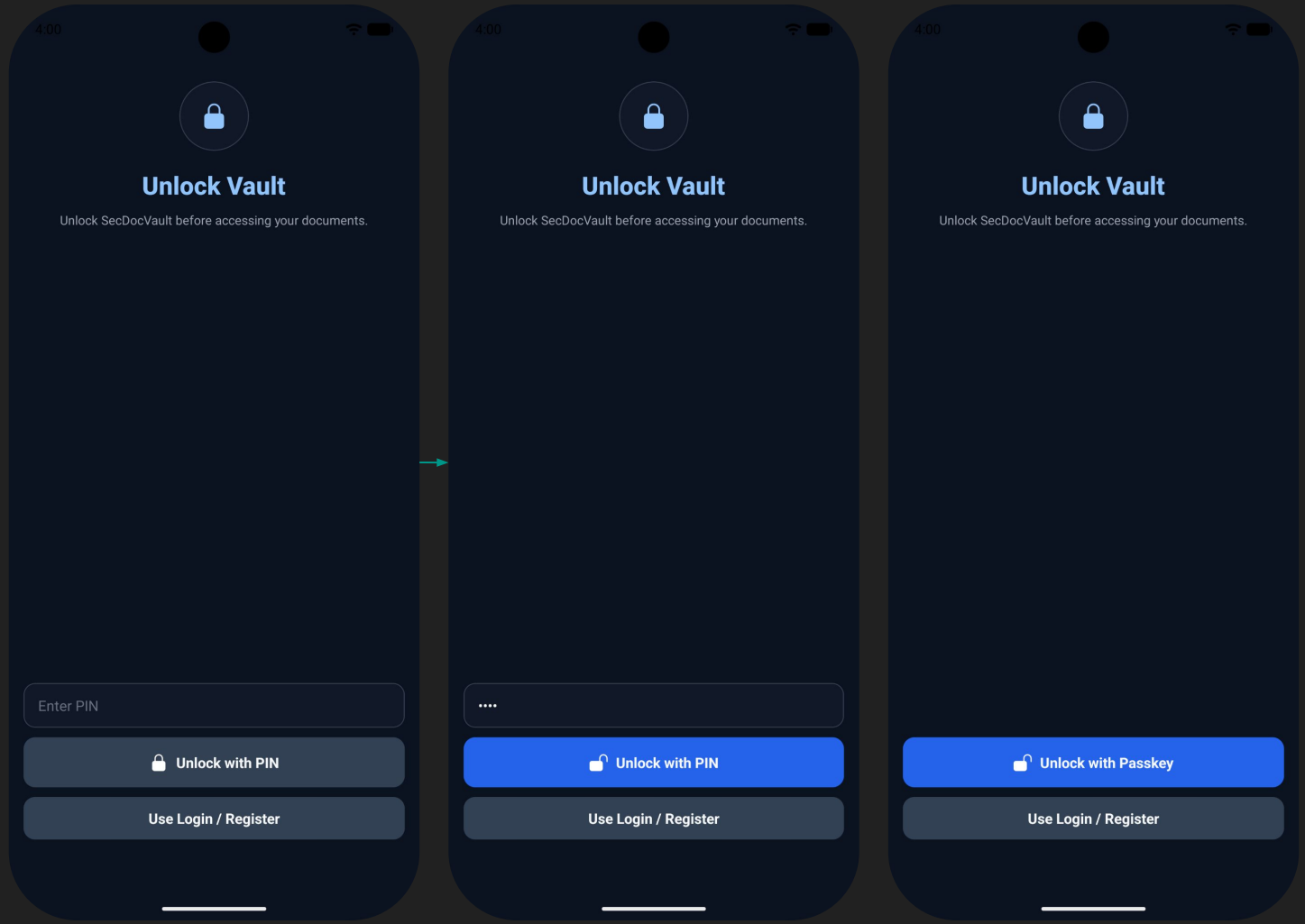




Finish Authentication Process with setting an Unlock Method

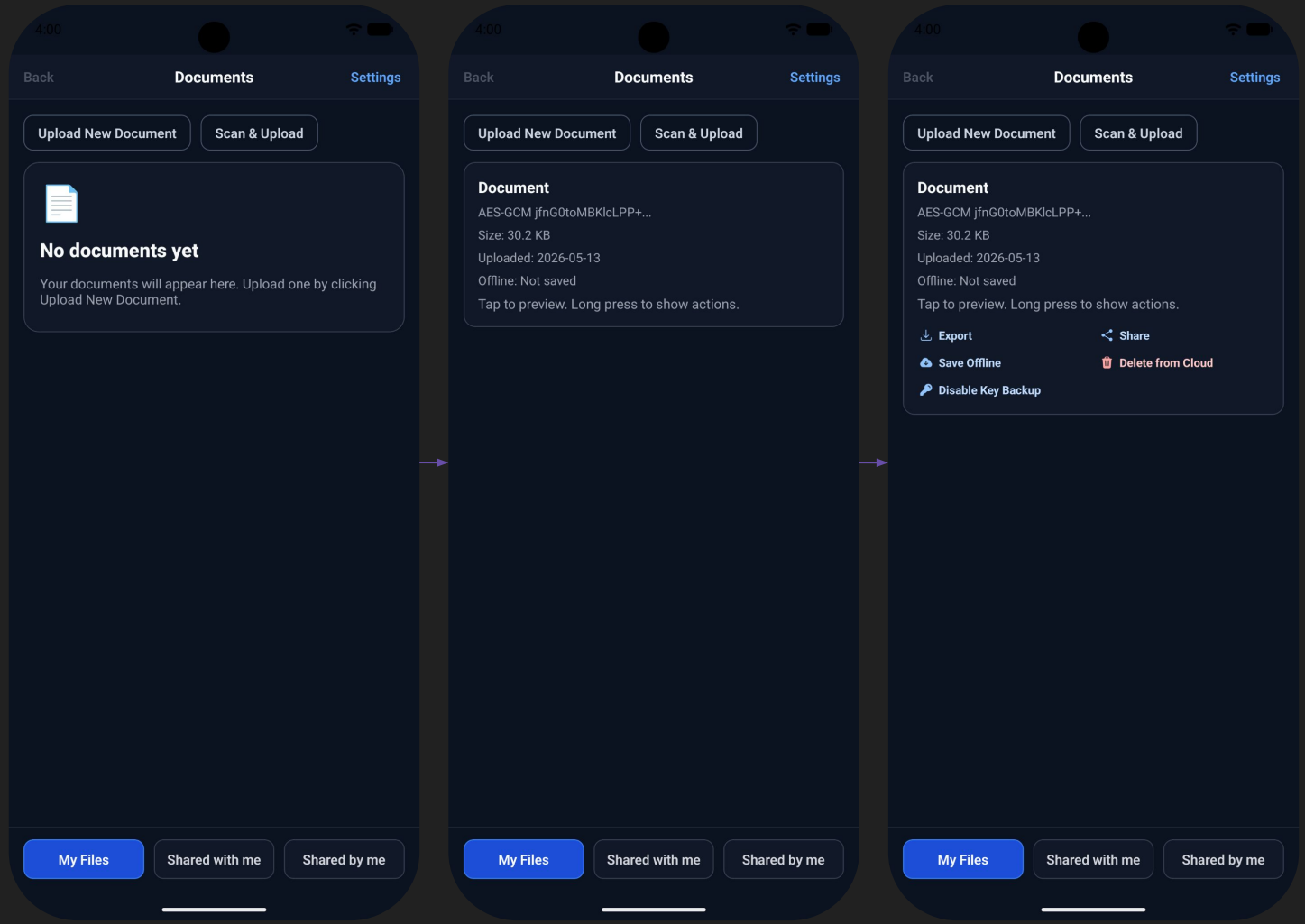
Authentication Process — Register or Login

Unlock Screen

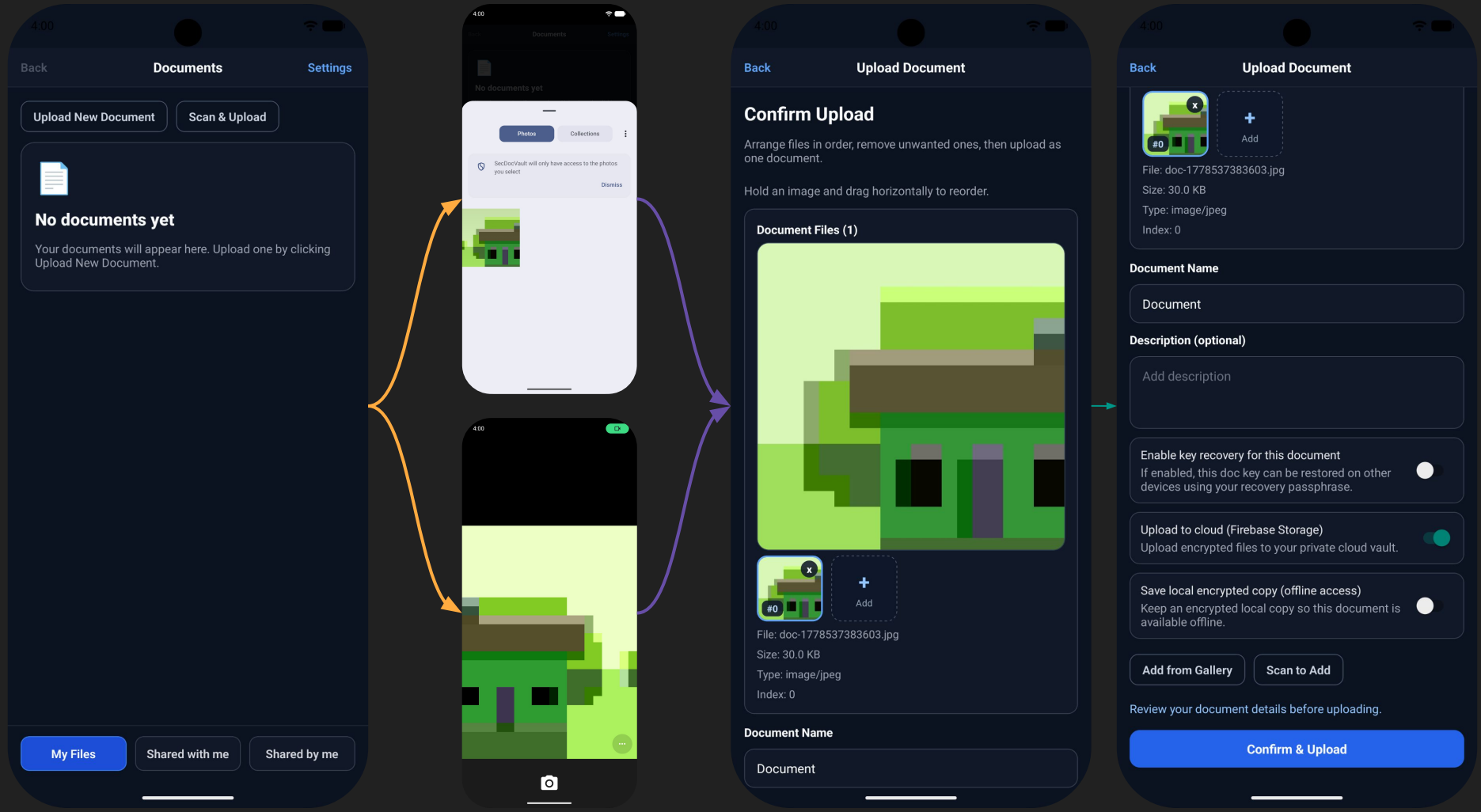


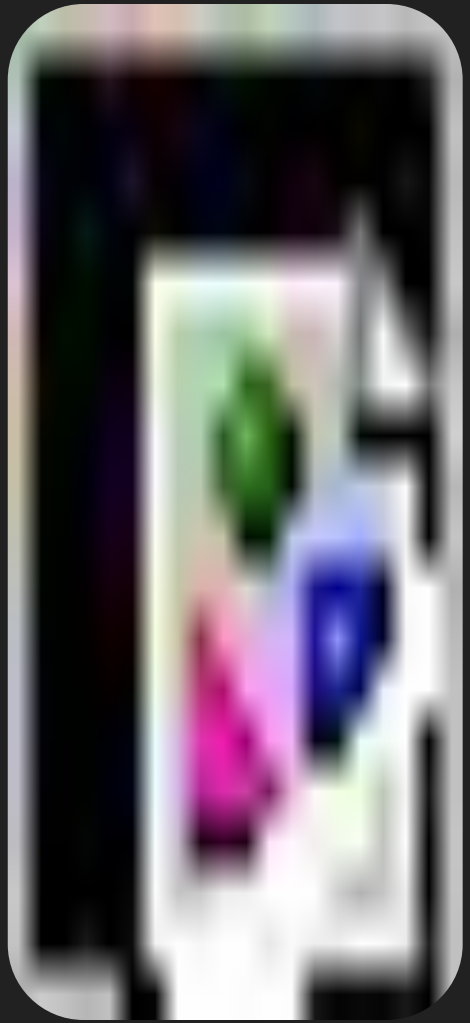
- App Exit
- Inactivity Auto-Lock

Home Screen


4:00
Back
Documents
Settings
Upload New Document
Scan & Upload
No documents yet
Your documents will appear here. Upload one by clicking Upload New Document.
My Files
Shared with me
Shared by me
4:00
Back
Documents
Settings
Upload New Document
Scan & Upload
Document
AES-GCM jfnG0toMBKlcLPP+...
Size: 30.2 KB
Uploaded: 2026-05-13
Offline: Not saved
Tap to preview. Long press to show actions.
My Files
Shared with me
Shared by me
4:00
Back
Documents
Settings
Upload New Document
Scan & Upload
Document
AES-GCM jfnG0toMBKlcLPP+...
Size: 30.2 KB
Uploaded: 2026-05-13
Offline: Not saved
Tap to preview. Long press to show actions.
Export
Share
Save Offline
Delete from Cloud
Disable Key Backup
My Files
Shared with me
Shared by me

Document Lifecycle


Back
Documents
Settings
Upload New Document
Scan & Upload
No documents yet
Your documents will appear here. Upload one by clicking Upload New Document.
My Files
Shared with me
Shared by me
Photos
Collections
Dismiss
Back
Upload Document
Confirm Upload
Arrange files in order, remove unwanted ones, then upload as one document.
Hold an image and drag horizontally to reorder.
Document Files (1)
#0
Add
File: doc-1778537383603.jpg
Size: 30.0 KB
Type: image/jpeg
Index: 0
Document Name
Document
Back
Upload Document
#0
Add
File: doc-1778537383603.jpg
Size: 30.0 KB
Type: image/jpeg
Index: 0
Document Name
Document
Description (optional)
Add description
Enable key recovery for this document
If enabled, this doc key can be restored on other devices using your recovery passphrase.
Upload to cloud (Firebase Storage)
Upload encrypted files to your private cloud vault.
Save local encrypted copy (offline access)
Keep an encrypted local copy so this document is available offline.
Add from Gallery
Scan to Add
Review your document details before uploading.
Confirm & Upload



Upload, Encryption and Persistence

1. File/files uploaded, name added
2. Generate random enc key
3. Encryption (repeat for each file)
    1. Convert file to base64
    2. Encrypt with enc key and IV using AES 256 GCM
    3. Save to storage as IV + cipher
    4. Create references array
4. Save enc key + docID to **local KeyStore**
5. Use derived passphrase from local keyStore with unique IV to encrypt random doc enc key.
6. Add array from step 5, encrypted key and IV from step 3 and metadata (name, owner, sharedWith) to a document in **docsVault collection**

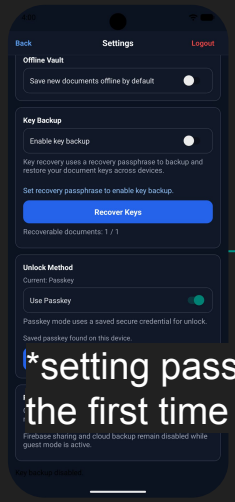


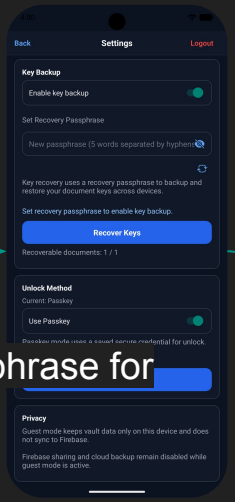
Decrypt, View and Export the Document

We also generate AES-GCM integrity tag to identify tampting

Backup Restoration

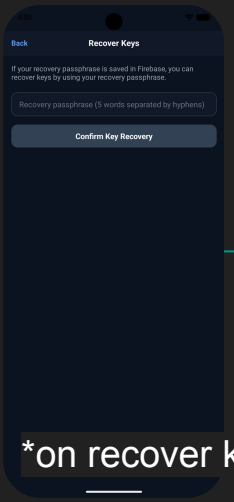
back to passphrase)

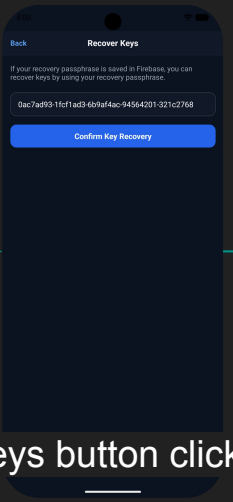


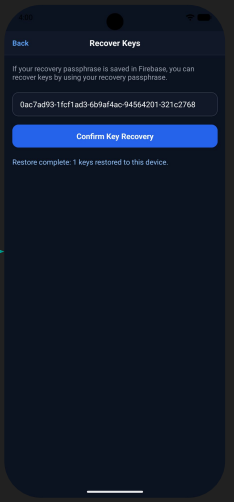


*setting passphrase for the first time

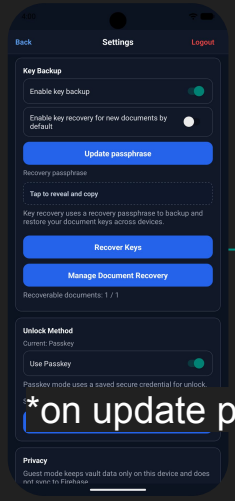
enable keys recovery and set a passphrase

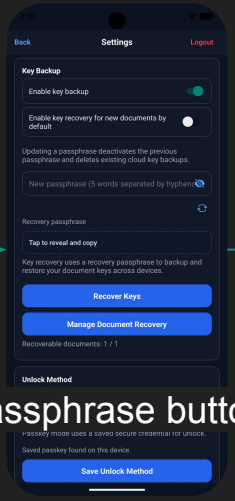


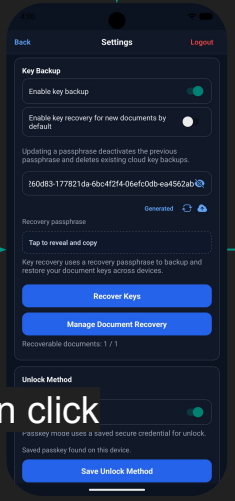


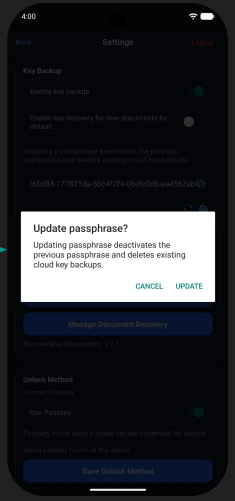


*on recover keys button click

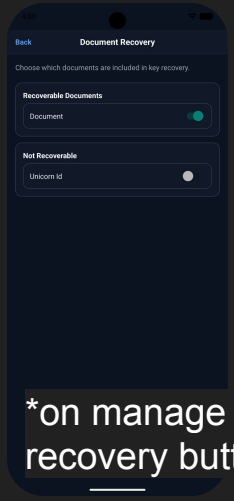








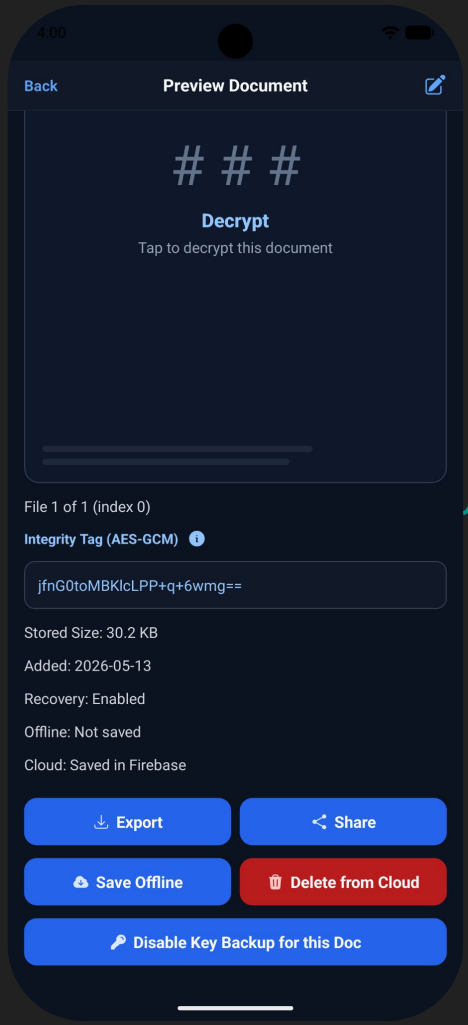
*on update passphrase button click

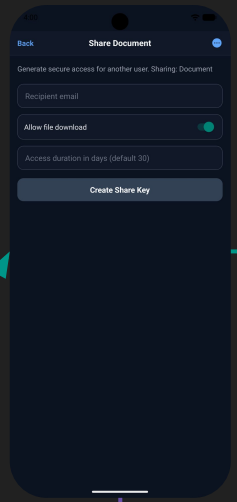


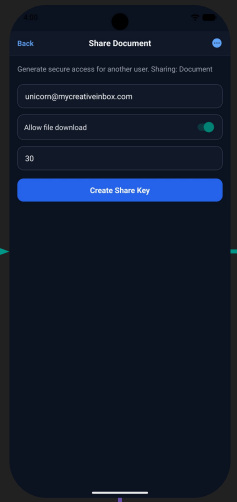
*on manage document recovery button click

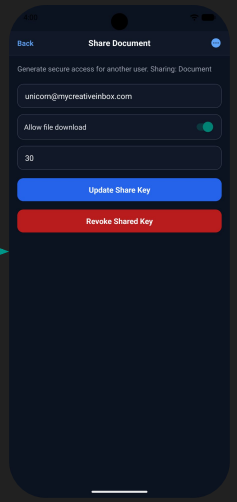
Sharing

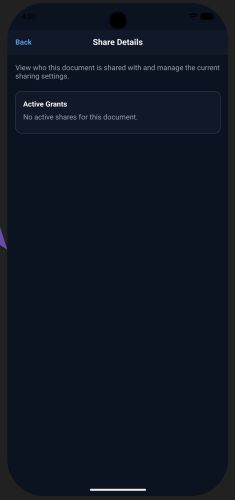
from ‘device 1’ and ‘device 2’

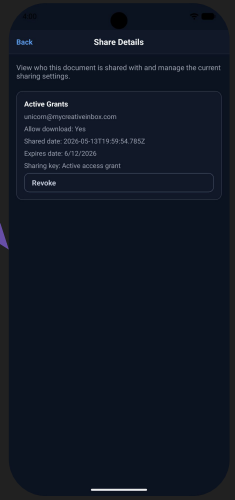

Back
Preview Document
# # #
Decrypt
Tap to decrypt this document
File 1 of 1 (index 0)
Integrity Tag (AES-GCM)
jfnG0toMBKlcLPP+q+6wmg==
Stored Size: 30.2 KB
Added: 2026-05-13
Recovery: Enabled
Offline: Not saved
Cloud: Saved in Firebase
Export
Share
Save Offline
Delete from Cloud
Disable Key Backup for this Doc

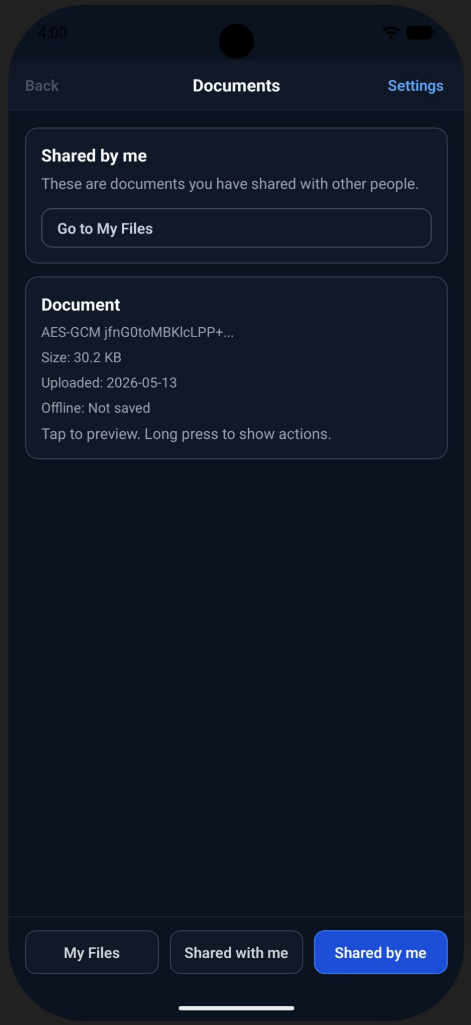

Back
Share Document
Generate secure access for another user. Sharing: Document
Recipient email
Allow file download
Access duration in days (default 30)
Create Share Key

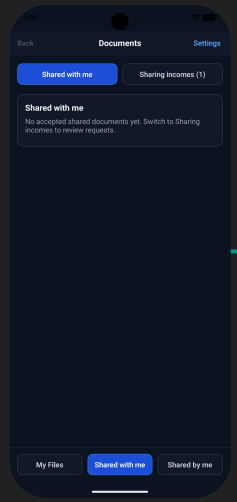

Back
Share Document
Generate secure access for another user. Sharing: Document
unicorn@mycreativeinbox.com
Allow file download
30
Create Share Key

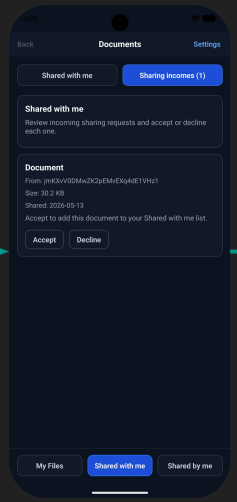

Back
Share Document
Generate secure access for another user. Sharing: Document
unicorn@mycreativeinbox.com
Allow file download
30
Update Share Key
Revoke Shared Key

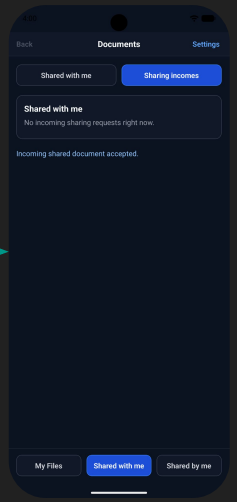

Back
Share Details
View who this document is shared with and manage the current sharing settings.
Active Grants
No active shares for this document.

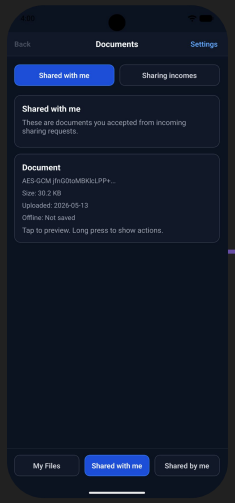

Back
Share Details
View who this document is shared with and manage the current sharing settings.
Active Grants
unicorn@mycreativeinbox.com
Allow download: Yes
Shared date: 2026-05-13T19:59:54.785Z
Expires date: 6/12/2026
Sharing key: Active access grant
Revoke

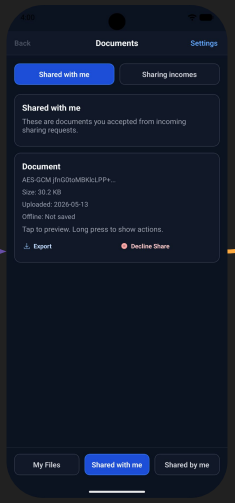

Back
Documents
Settings
Shared by me
These are documents you have shared with other people.
Go to My Files
Document
AES-GCM jfnG0toMBKlcLPP+...
Size: 30.2 KB
Uploaded: 2026-05-13
Offline: Not saved
Tap to preview. Long press to show actions.
My Files
Shared with me
Shared by me

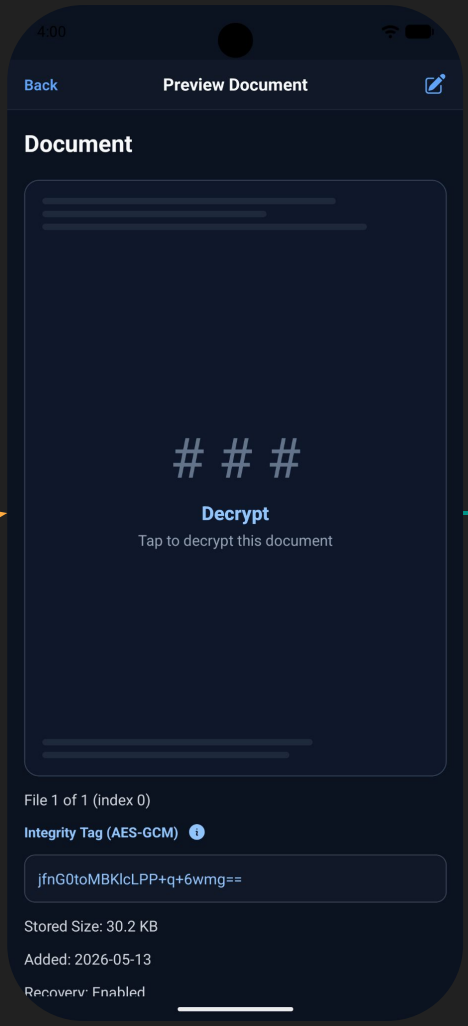

Back
Documents
Settings
Shared with me
Sharing incomes (1)
Shared with me
No accepted shared documents yet. Switch to Sharing incomes to review requests.
My Files
Shared with me
Shared by me


Back
Documents
Settings
Shared with me
Sharing incomes (1)
Shared with me
Review incoming sharing requests and accept or decline each one.
Document
From: jmKXvV0DMwZK2pEMvEXq4dE1VHz1
Size: 30.2 KB
Shared: 2026-05-13
Accept to add this document to your Shared with me list.
Accept
Decline
My Files
Shared with me
Shared by me


Back
Documents
Settings
Shared with me
Sharing incomes
Shared with me
No incoming sharing requests right now.
Incoming shared document accepted.
My Files
Shared with me
Shared by me

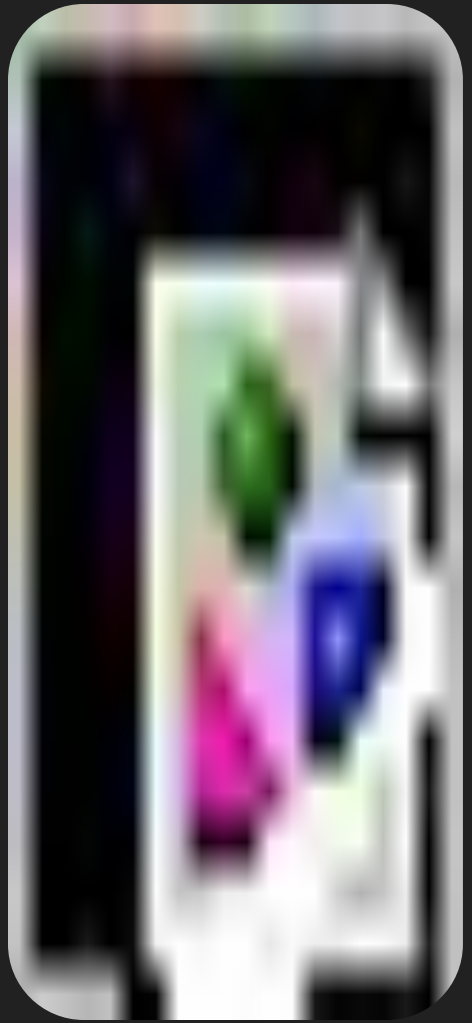

Back
Documents
Settings
Shared with me
Sharing incomes
Shared with me
These are documents you accepted from incoming sharing requests.
Document
AES-GCM jfnG0toMBKlcLPP+...
Size: 30.2 KB
Uploaded: 2026-05-13
Offline: Not saved
Tap to preview. Long press to show actions.
My Files
Shared with me
Shared by me

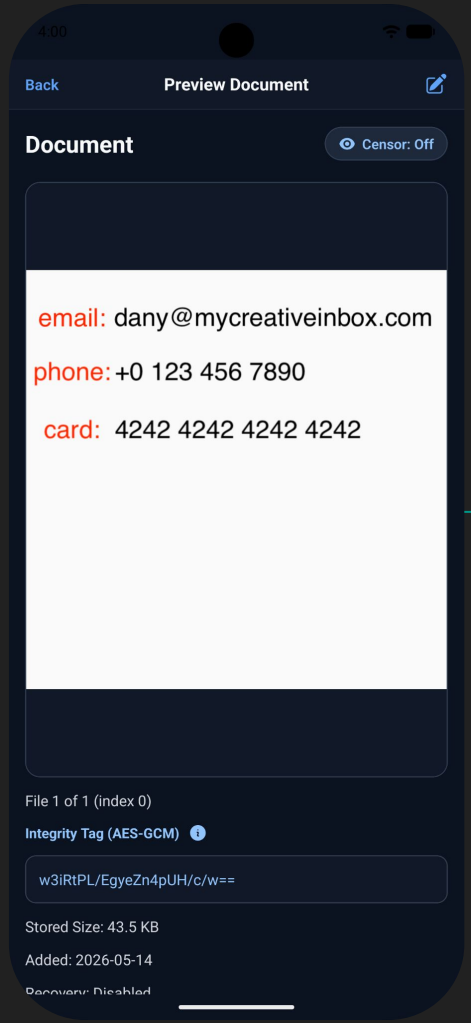

Back
Documents
Settings
Shared with me
Sharing incomes
Shared with me
These are documents you accepted from incoming sharing requests.
Document
AES-GCM jfnG0toMBKlcLPP+...
Size: 30.2 KB
Uploaded: 2026-05-13
Offline: Not saved
Tap to preview. Long press to show actions.
Export
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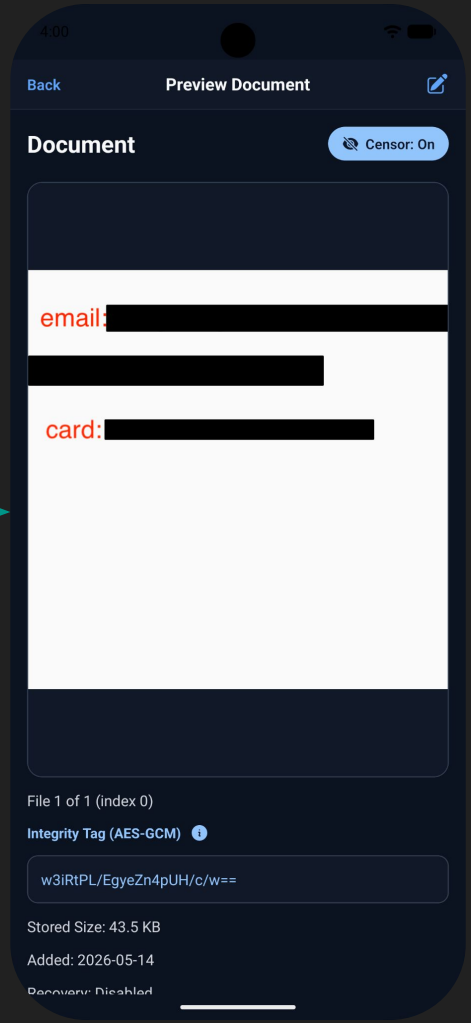

4:00
Back
Preview Document
Document
# # #
Decrypt
Tap to decrypt this document
File 1 of 1 (index 0)
Integrity Tag (AES-GCM)
jfnG0toMBKlcLPP+q+6wmg==
Stored Size: 30.2 KB
Added: 2026-05-13
Recovery: Enabled


4:00
Back
Preview Document
Document
Censor: Off
File 1 of 1 (index 0)
Integrity Tag (AES-GCM)
jfnG0toMBKlcLPP+q+6wmg==
Stored Size: 30.2 KB
Added: 2026-05-13
Recovery: Enabled

Other Features







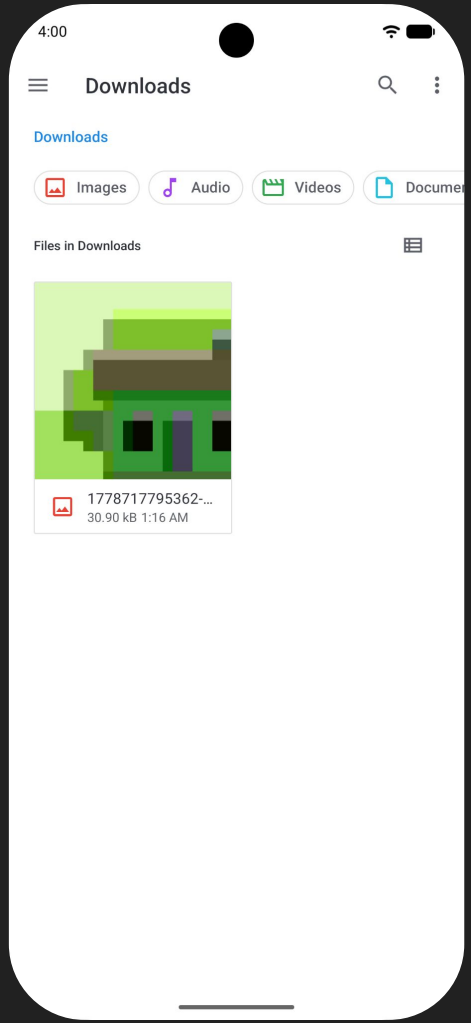
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Document Censoring

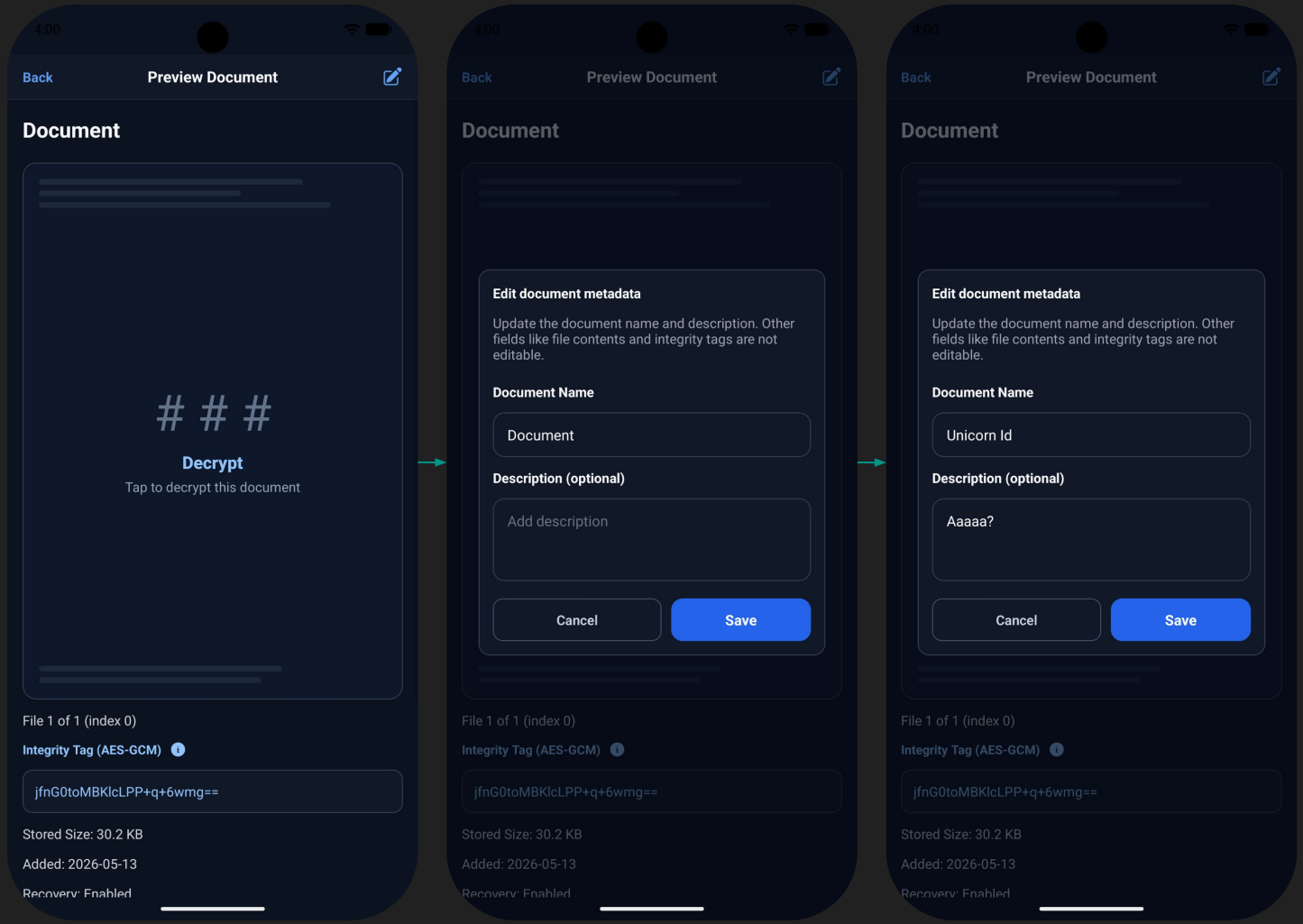
Export Document File



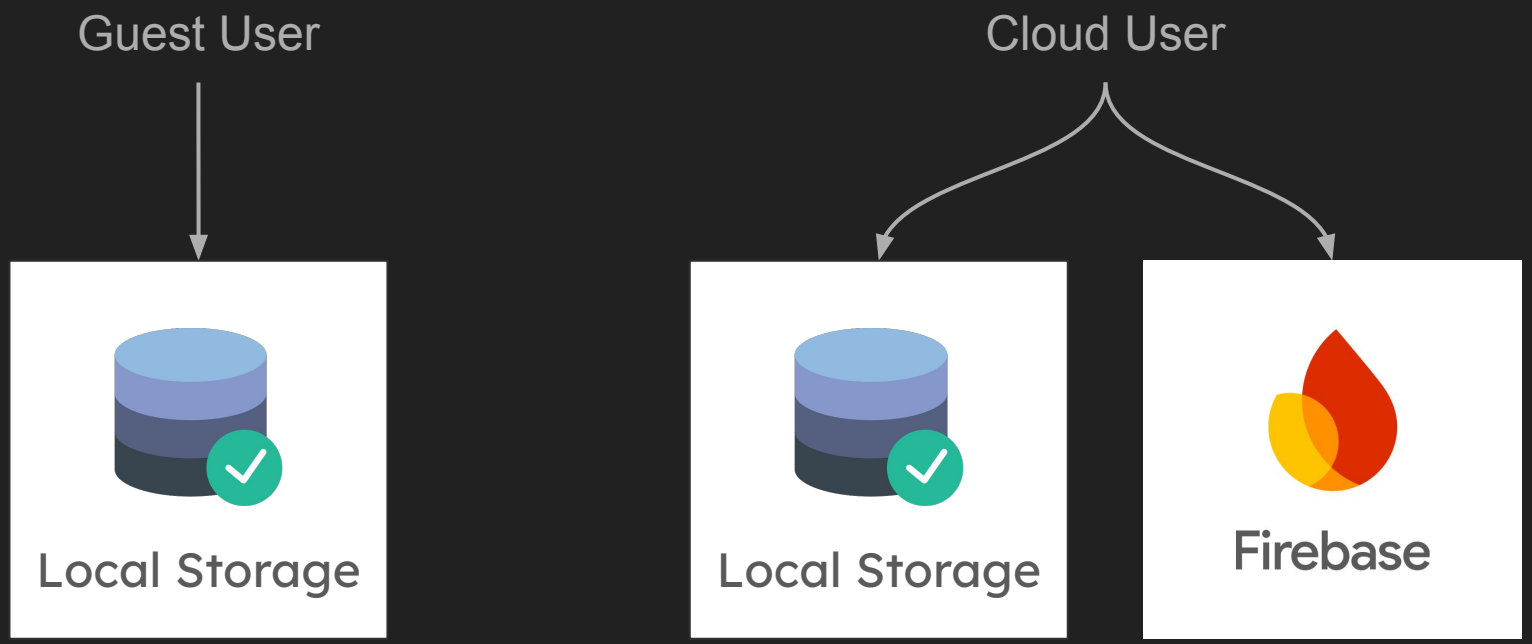




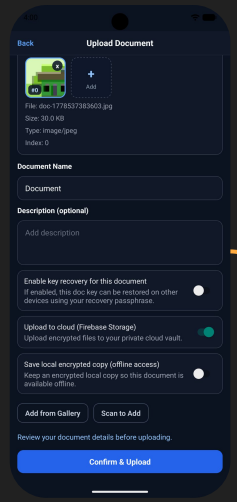
Document Metadata Editing

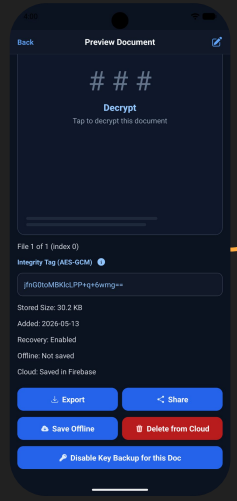


Storage Mode Selection

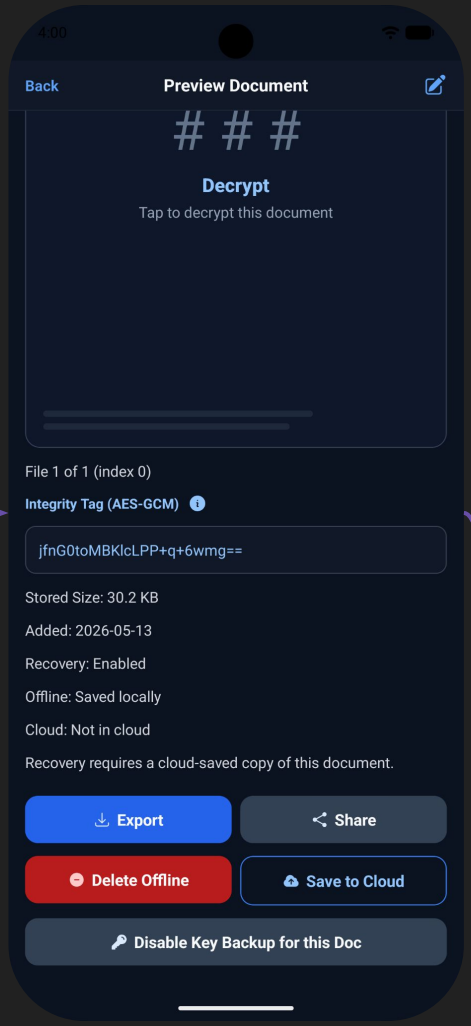


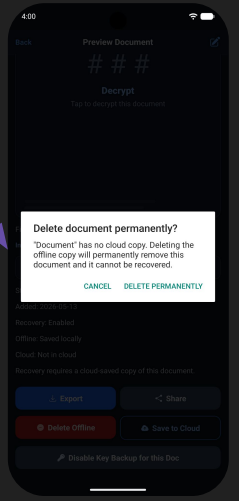
Save/Delete Local (Offline Access)

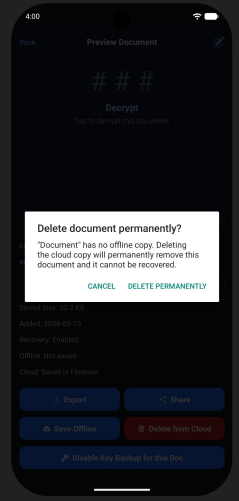






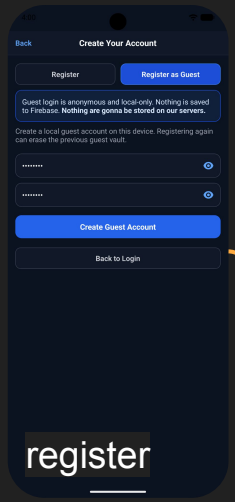


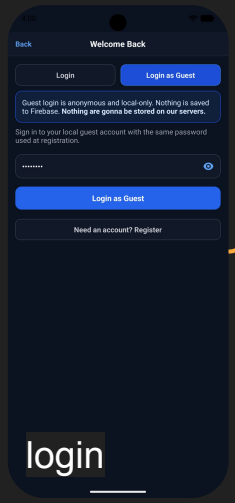


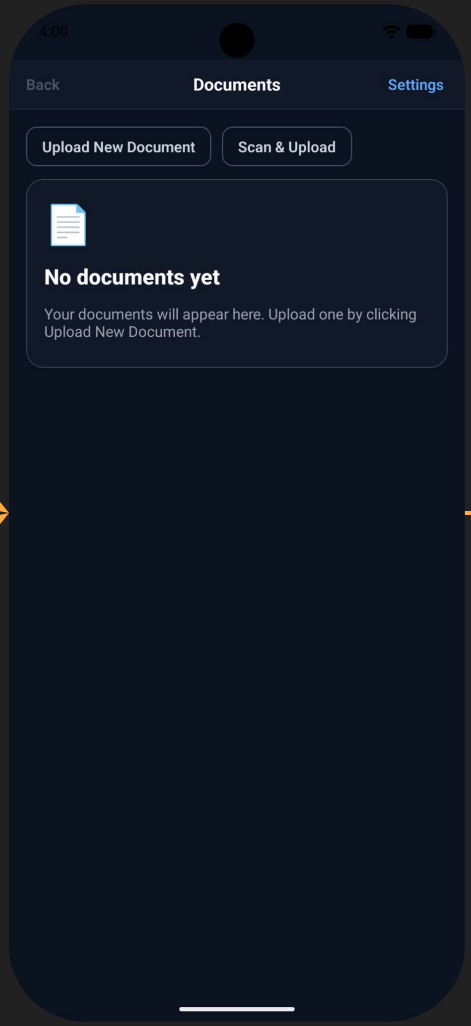


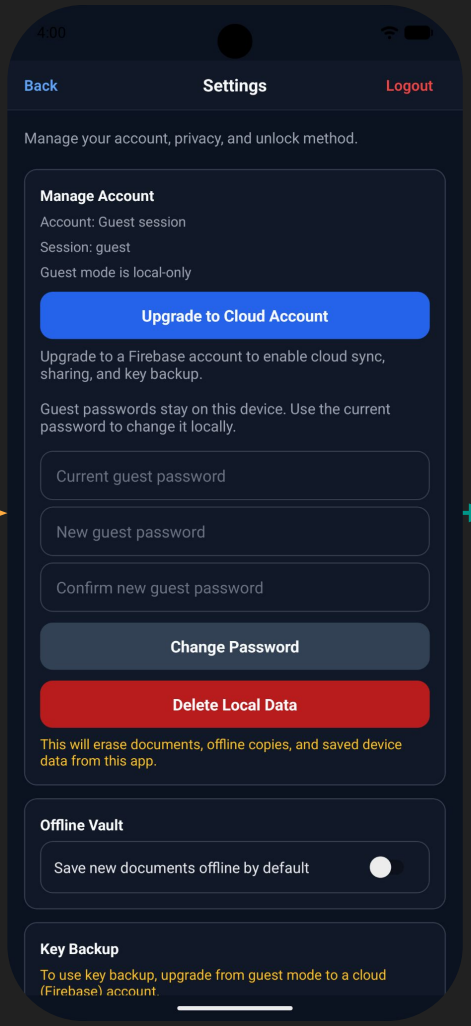
Guest (Anonymous) Account

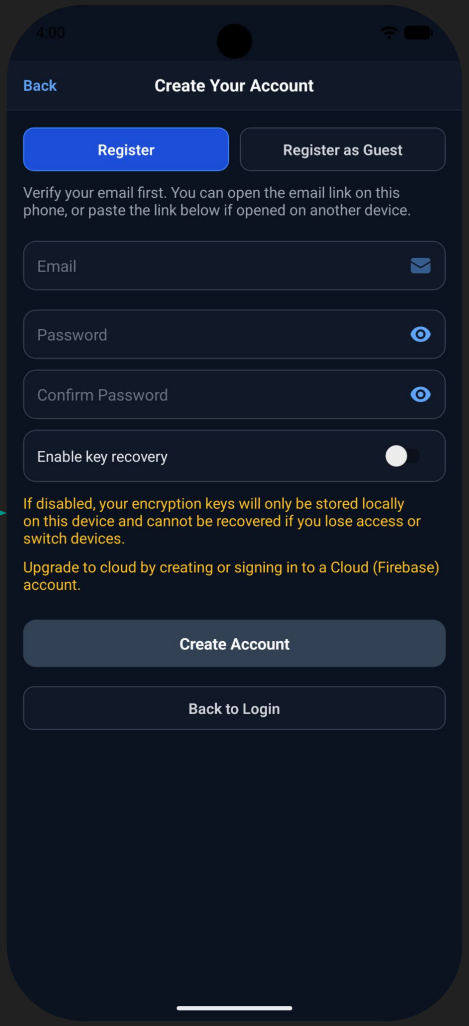
let's come back to authentication


Back
Create Your Account
Register
Register as Guest
Guest login is anonymous and local-only. Nothing is saved to Firebase. Nothing are gonna be stored on our servers.
Create a local guest account on this device. Registering again can erase the previous guest vault.
Create Guest Account
Back to Login
register


Back
Welcome Back
Login
Login as Guest
Guest login is anonymous and local-only. Nothing is saved to Firebase. Nothing are gonna be stored on our servers.
Sign in to your local guest account with the same password used at registration.
Login as Guest
Need an account? Register
login


Back
Documents
Settings
Upload New Document
Scan & Upload
No documents yet
Your documents will appear here. Upload one by clicking Upload New Document.


Back
Settings
Logout
Manage your account, privacy, and unlock method.
Manage Account
Account: Guest session
Session: guest
Guest mode is local-only
Upgrade to Cloud Account
Upgrade to a Firebase account to enable cloud sync, sharing, and key backup.
Guest passwords stay on this device. Use the current password to change it locally.
Current guest password
New guest password
Confirm new guest password
Change Password
Delete Local Data
This will erase documents, offline copies, and saved device data from this app.
Offline Vault
Save new documents offline by default
Key Backup
To use key backup, upgrade from guest mode to a cloud (Firebase) account.


Back
Create Your Account
Register
Register as Guest
Verify your email first. You can open the email link on this phone, or paste the link below if opened on another device.
Email
Password
Confirm Password
Enable key recovery
If disabled, your encryption keys will only be stored locally on this device and cannot be recovered if you lose access or switch devices.
Upgrade to cloud by creating or signing in to a Cloud (Firebase) account.
Create Account
Back to Login

Background Privacy Screen

What user sees when app is in the background



How screenshot looks like



Test Coverage

```
Test Suites: 80 passed, 80 total
Tests:       663 passed, 663 total
Snapshots:   0 total
Time:        3.769 s, estimated 4 s
Ran all test suites.
```

66.62% Statements `2907/4363` **56.62%** Branches `1800/3179` **65.74%** Functions `568/864` **67.44%** Lines `2817/4177`

File	Statements		Branches		Functions		Lines	
SecDocVault	100%	2/2	100%	2/2	100%	1/1	100%	2/2
SecDocVault/src/app/components	96.61%	57/59	94.3%	116/123	100%	23/23	96.61%	57/59
SecDocVault/src/app/controllers	79.34%	315/397	60.55%	109/180	80.2%	77/96	79.79%	312/391
SecDocVault/src/app/hooks	72.48%	685/945	58.84%	296/503	69.18%	110/159	73.35%	669/912
SecDocVault/src/app/navigation	97.36%	37/38	97.05%	33/34	100%	6/6	97.36%	37/38
SecDocVault/src/components	97.05%	33/34	89.87%	71/79	100%	12/12	97.05%	33/34
SecDocVault/src/context	43.1%	294/682	32.28%	92/285	52.87%	46/87	43.41%	287/661
SecDocVault/src/firebase	100%	4/4	100%	0/0	100%	0/0	100%	4/4
SecDocVault/src/screens	47.47%	319/672	46.27%	441/953	38.98%	92/236	50.24%	302/601
SecDocVault/src/services	32.14%	54/168	21.68%	18/83	68.42%	13/19	31.73%	53/167
SecDocVault/src/services/censor	60.07%	170/283	29.41%	65/221	58.06%	18/31	60.14%	163/271
SecDocVault/src/services/crypto	50.79%	96/189	45.56%	36/79	52.5%	21/40	50%	92/184
SecDocVault/src/services/documentVault	94.2%	796/845	81.26%	503/619	96.52%	139/144	94.18%	761/808
SecDocVault/src/storage	100%	37/37	100%	18/18	100%	10/10	100%	37/37
SecDocVault/src/theme	100%	2/2	100%	0/0	100%	0/0	100%	2/2
SecDocVault/src/theme/styleComponents	100%	1/1	100%	0/0	100%	0/0	100%	1/1
SecDocVault/src/theme/styleSections	100%	5/5	100%	0/0	100%	0/0	100%	5/5

Thanks for attention!

questions?